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Pattern matching method

Abstract

The method includes: determine a first integrated value by integrating measured values of widths of reference patterns (**210**A) belonging to a first group; determine a second integrated value by integrating measured values of widths of reference patterns (**210**B) belonging to a second group; performing second matching between patterns on an image of a second region and corresponding CAD patterns; determining a third integrated value by integrating measured values of widths of patterns (**220**A) belonging to a first group; determining a fourth integrated value by integrating measured values of widths of patterns (**220**B) belonging to a second group; and determining that the second matching has been performed correctly when the magnitude relationship between the third integrated value and the fourth integrated value coincides with the magnitude relationship between the first integrated value and the second integrated value.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is a 35 U.S.C. § 371 filing of International Application No. PCT/JP2021/015017 filed Apr. 9, 2021, which claims the benefit of priority to Japanese Patent Application No. 2020-074033 filed Apr. 17, 2020, each of which is incorporated herein by reference in its entirety.
TECHNICAL FIELD
(2) The present invention relates to a method of matching a pattern formed on a surface of a workpiece, such as a wafer or a glass substrate, used for manufacturing semiconductor devices with a CAD pattern created from pattern design data.
BACKGROUND ART
(3) A die-to-database method is a pattern matching method for matching a pattern formed on a surface of a workpiece, such as a wafer or a glass substrate, with a CAD pattern created from pattern design data. More specifically, the die-to-database method includes obtaining coordinates of an area to be inspected from design data, moving a stage on which the workpiece is placed to the coordinates, generating an image of a pattern on the workpiece by electron-beam irradiation, superimposing the image and a CAD pattern created from the design data, producing a gray-level profile of the image within a set range starting from an edge of the CAD pattern, determining an edge of the pattern on the image based on the gray-level profile, and determining a matching position that minimize a bias value between the determined edge position and the edge of the corresponding CAD pattern (see, for example, Patent Document 1).
CITATION LIST
Patent Literature
(4) Patent document 1: Japanese laid-open patent publication No. 5-324836 Patent document 2: Japanese laid-open patent publication No. 2012-519391
SUMMARY OF INVENTION
Technical Problem
(5) Double patterning, which is capable of forming a highly integrated circuit having a narrower pattern interval, has recently been attracting attention. The double patterning is a technique of forming a first pattern and a second pattern separately, which are arranged alternately, in two steps. However, in certain areas, a variation in pattern dimension (CD, or Critical Dimension) is likely to occur. Therefore, in such an area, it is necessary to acquire accurate statistical data of CD (Critical Dimension) in order to optimize the process and monitor the process fluctuation.
(6) Patent Document 2 discloses a technique of measuring CD of a pattern 1 and a pattern 2 using only features of a pattern profile of a SEM image. However, with this technique, it is difficult to analyze a problem in design or process because information that can be added to design

information other than the measured value of CD cannot be obtained.
(7) The conventional die-to-database method can realize operations of directly specifying an area that is expected to be greatly affected by process fluctuation of the double patterning on the CAD coordinate system in the design data to obtain CD statistical data, and evaluating a CD variation in a memory cell. Furthermore, the die-to-database method can measure the CD while correcting effects of magnification change and rotation of a SEM image using the design data, and can therefore obtain a difference between the CDs and a large amount of accurate design values. In addition, since additional information, such as peripheral pattern information, can be obtained from the design data, the die-to-database method is considered to be most suitable for CD measurement data analysis for the purpose of photomasks and changing of process parameters.
(8) Usually, at an end of a memory cell, a pattern is easily deformed greatly due to influence of the light proximity effect, etc. Therefore, in order to acquire the data of the CD fluctuation caused by the double patterning, it is ideal to perform the CD measurement at the central part of the memory cell. However, in the central part of the memory cell, it is difficult to judge whether the pattern matching is correct or not. For example, as shown in FIG. **16**, although repeating patterns **505** on a SEM image **500** match CAD patterns **510** on design data, a pitch shift may occur.
(9) Therefore, the present invention provides a pattern matching method capable of achieving correct matching between a pattern formed by multi-patterning and a corresponding CAD pattern.
Solution to Problem
(10) In an embodiment, there is provided a pattern matching method comprising: generating an image of a first region in a pattern region containing patterns formed by multi-patterning; performing first matching between a plurality of reference patterns on the image of the first region and a plurality of first CAD patterns that have been classified in advance into a first group and a second group according to layer information; classifying the plurality of reference patterns into the first group and the second group according to the layer classification of the first CAD patterns: measuring widths of the plurality of reference patterns belonging to the first group and widths of the plurality of reference patterns belonging to the second group: determining a first integrated value by integrating measured values of the widths of the plurality of reference patterns belonging to the first group; determining a second integrated value by integrating measured values of the widths of the plurality of reference patterns belonging to the second group: determining a magnitude relationship between the first integrated value and the second integrated value: generating an image of a second region in the pattern region; performing second matching between a plurality of patterns on the image of the second region and a plurality of second CAD patterns that have been classified in advance into a first group and a second group according to layer information; classifying the plurality of patterns on the image of the second region into the first group and the second group according to the layer classification of the second CAD patterns; measuring widths of the plurality of patterns belonging to the first group and widths of the plurality of patterns belonging to the second group: determining a third integrated value by integrating measured values of the widths of the plurality of patterns belonging to the first group; determining a fourth integrated value by integrating measured values of the widths of the plurality of patterns belonging to the second group; determining a magnitude relationship between the third integrated value and the fourth integrated value; determining that the second matching has been performed correctly when the magnitude relationship between the third integrated value and the fourth integrated value coincides with the magnitude relationship between the first integrated value and the second integrated value.
(11) In an embodiment, the first region is an edge region including an edge of the pattern region, and the second region is within the pattern region and is located more inwardly than the first region.
(12) In an embodiment, the patterns formed by the multi-patterning are repeating patterns.
(13) In an embodiment, the pattern matching method further comprises: shifting the plurality of

patterns on the image of the second region relative to the plurality of second CAD patterns by one pitch when the magnitude relationship between the third integrated value and the fourth integrated value does not coincide with the magnitude relationship between the first integrated value and the second integrated value; and then performing the second matching again.

(14) In an embodiment, an absolute value of a difference between the first integrated value and the second integrated value and an absolute value of a difference between the third integrated value and the fourth integrated value are larger than a predetermined value.

(15) In an embodiment, there is provided a pattern matching method comprising: generating an image of a first region in a pattern region containing patterns formed by multi-patterning; performing first matching between a plurality of reference patterns on the image of the first region and a plurality of first CAD patterns that have been classified in advance into a first group and a second group according to layer information: classifying the plurality of reference patterns into the first group and the second group according to the layer classification of the first CAD patterns; calculating slopes of brightness profiles of edges of the plurality of reference patterns belonging to the first group and slopes of brightness profiles of edges of the plurality of reference patterns belonging to the second group; determining a first integrated value by integrating calculated values of the slopes of the brightness profiles of the edges of the plurality of reference patterns belonging to the first group; determining a second integrated value by integrating calculated values of the slopes of the brightness profiles of the edges of the plurality of reference patterns belonging to the second group: determining a magnitude relationship between the first integrated value and the second integrated value; generating an image of a second region in the pattern region; performing second matching between a plurality of patterns on the image of the second region and a plurality of second CAD patterns that have been classified in advance into a first group and a second group according to layer information; classifying the plurality of patterns on the image of the second region into the first group and the second group according to the layer classification of the second CAD patterns; calculating slopes of brightness profiles of edges of the plurality of patterns belonging to the first group and slopes of brightness profiles of edges of the plurality of patterns belonging to the second group; determining a third integrated value by integrating calculated values of the slopes of the brightness profiles of the edges of the plurality of patterns belonging to the first group; determining a fourth integrated value by integrating calculated values of the slopes of the brightness profiles of the edges of the plurality of patterns belonging to the second group; determining a magnitude relationship between the third integrated value and the fourth integrated value: determining that the second matching has been performed correctly when the magnitude relationship between the third integrated value and the fourth integrated value coincides with the magnitude relationship between the first integrated value and the second integrated value.

(16) In an embodiment, the first region is an edge region including an edge of the pattern region, and the second region is within the pattern region and is located more inwardly than the first region.

(17) In an embodiment, the patterns formed by the multi-patterning are repeating patterns.

(18) In an embodiment, the pattern matching method further comprises: shifting the plurality of patterns on the image of the second region relative to the plurality of second CAD patterns by one pitch when the magnitude relationship between the third integrated value and the fourth integrated value does not coincide with the magnitude relationship between the first integrated value and the second integrated value; and then performing the second matching again.

(19) In an embodiment, an absolute value of a difference between the first integrated value and the second integrated value and an absolute value of a difference between the third integrated value and the fourth integrated value are larger than a predetermined value.

Advantageous Effects of Invention

(20) According to the present invention, accurate pattern matching in the second region can be guaranteed by referring to the magnitude relationship of the pattern width in the first region. In

particular, according to the present invention, the measured value of the pattern width can be used for optimizing the process parameters. In addition, it is possible to monitor the pattern width in an area where the process margin is small.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a schematic diagram showing an embodiment of an image generating apparatus.

(2) FIG. **2** is a schematic diagram showing a pattern region formed on a wafer.

(3) FIG. **3** is a schematic diagram showing an example of an image of a first region.

(4) FIG. **4** is a diagram explaining first matching of a plurality of reference patterns on the image of the first region and a plurality of corresponding CAD patterns.

(5) FIG. **5** is a diagram illustrating a process of classifying a plurality of reference patterns on an SEM image into a first group and a second group.

(6) FIG. **6** is a graph showing a relationship between the number of integrated widths of reference patterns and first and second integrated values.

(7) FIG. **7** is a schematic diagram showing another example of patterns in the first region.

(8) FIG. **8** is a schematic diagram showing an example of an image of a second region.

(9) FIG. **9** is a diagram illustrating second matching between a plurality of patterns on the image of the second region and a plurality of corresponding CAD patterns.

(10) FIG. **10** is a diagram illustrating a process of classifying a plurality of patterns on an SEM image into a first group and a second group.

(11) FIG. **11** is a flowchart explaining an embodiment of a pattern matching method.

(12) FIG. **12** is a schematic diagram showing another example of an image of a first region.

(13) FIG. **13** is a schematic diagram showing another example of an image in a second region.

(14) FIG. **14** is a diagram showing an example of a brightness profile of a pattern on an image.

(15) FIG. **15** is a flowchart explaining an embodiment of a pattern matching method.

(16) FIG. **16** is a diagram explaining a problem that may occur in conventional pattern matching.

DESCRIPTION OF EMBODIMENTS

(17) Hereinafter, embodiments of the present invention will be described with reference to the drawings.

(18) FIG. **1** is a schematic diagram showing an embodiment of an image generation apparatus. As shown in FIG. **1**, the image generating apparatus includes a scanning electron microscope **50** and an arithmetic system **150**. The scanning electron microscope **50** is coupled to the arithmetic system **150**, and operations of the scanning electron microscope **50** are controlled by the arithmetic system **150**.

(19) The arithmetic system **150** includes a memory **162** storing a database **161** and programs therein, a processor **163** configured to perform arithmetic operations according to instructions included in the programs, and a display screen **165** configured to display an image and a GUI (graphical user interface). The processor **163** includes a CPU (central processing unit) or a GPU (graphic processing unit) that performs arithmetic operations according to instructions included in the programs stored in the memory **162**. The memory **162** includes a main memory (e.g., random access memory) to which the processor **163** is accessible and an auxiliary memory (e.g., hard disk drive or solid state drive) for storing data and the programs.

(20) The arithmetic system **150** includes at least one computer. For example, the arithmetic system **150** may be an edge server coupled to the scanning electron microscope **50** by a communication line, or may be a cloud server coupled to the scanning electron microscope **50** by a communication network, such as the Internet or a local network. The arithmetic system **150** may be a fog computing device (gateway, fog server, router, etc.) installed in a network coupled to the scanning

electron microscope **50**. The arithmetic system **150** may be a combination of a plurality of servers. For example, the arithmetic system **150** may be a combination of an edge server and a cloud server coupled to each other by a communication network, such as the Internet or a local network. In another example, the arithmetic system **150** may include a plurality of servers (computers) that are not coupled by a network.

(21) The scanning electron microscope **50** includes an electron gun **111** configured to emit an electron beam composed of primary electrons (charged particles), a converging lens **112** configured to converge the electron beam emitted by the electron gun **111**, and an X deflector **113** configured to deflect the electron beam in an X direction, a Y deflector **114** configured to deflect the electron beam in a Y direction, and an objective lens **115** configured to focus the electron beam on a wafer **124**, which is an example of a workpiece. Configuration of the electron gun **111** is not particularly limited. For example, a field-emitter type electron gun, a semiconductor-photocathode type electron gun, etc. can be used as the electron gun **111**.

(22) The converging lens **112** and the objective lens **115** are coupled to a lens controller **116**, and operations of the converging lens **112** and the objective lens **115** are controlled by the lens controller **116**. The lens controller **116** is coupled to the arithmetic system **150**. The X deflector **113** and the Y deflector **114** are coupled to a deflection controller **117**, and deflecting operations of the X deflector **113** and the Y deflector **114** are controlled by the deflection controller **117**. The deflection controller **117** is also coupled to the arithmetic system **150** as well. A secondary electron detector **130** and a backscattered electron detector **131** are coupled to an image acquisition device **118**. The image acquisition device **118** is configured to convert output signals of the secondary electron detector **130** and the backscattered electron detector **131** into image(s). The image acquisition device **118** is also coupled to the arithmetic system **150** as well.

(23) A specimen stage **121** arranged in a specimen chamber **120** is coupled to a stage controller **122**, and a position of the specimen stage **121** is controlled by the stage controller **122**. The stage controller **122** is coupled to the arithmetic system **150**. A transfer device **140** for transporting the wafer **124** onto the specimen stage **121** in the specimen chamber **120** is also coupled to the arithmetic system **150**.

(24) The electron beam emitted by the electron gun **111** is converged by the converging lens **112** and then focused by the objective lens **115** on a surface of the wafer **124** while the electron beam is deflected by the X deflector **113** and the Y deflector **114**. When the wafer **124** is irradiated with the primary electrons of the electron beam, the secondary electrons and backscattered electrons are emitted from the wafer **124**. The secondary electrons are detected by the secondary electron detector **130**, and the backscattered electrons are detected by the backscattered electron detector **131**. Signals of the detected secondary electrons and signals of the detected backscattered electrons are input to the image acquisition device **118** and converted into image(s). The image is transmitted to the arithmetic system **150**.

(25) Design data of patterns formed on the wafer **124** is stored in advance in the memory **162**. The design data includes pattern design information, such as coordinates of vertices of each pattern formed on the wafer **124**, a position, a shape and a size of each pattern, and the number of a layer to which each pattern belongs. The database **161** is constructed in the memory **162**. The design data of patterns is stored in advance in the database **161**. The arithmetic system **150** can read out the design data of each pattern from the database **161** stored in the memory **162**.

(26) Next, an embodiment of a method of matching between a pattern on an image generated by the scanning electron microscope **50** and a corresponding CAD pattern on the design data will be described. In the following descriptions, an image generated by the scanning electron microscope **50** may be referred to as a SEM image. The pattern of the wafer **124** has been manufactured based on the design data (also referred to as CAD data). CAD is an abbreviation for computer-aided design.

(27) The design data includes design information of the pattern formed on the wafer **124**, and

specifically, includes design information, such as coordinates of vertices of the pattern, a position, a shape and a size of the pattern, and the number of a layer to which the pattern belongs. The CAD pattern in the design data is a virtual pattern defined by design information of a pattern included in the design data. In the following description, a pattern already formed on the wafer **124** may be referred to as a real pattern.

(28) FIG. **2** is a schematic view showing a pattern region **200** formed on the wafer **124**. Patterns formed in the pattern region **200** are real patterns formed by multi-patterning, such as double patterning or quadruple patterning. In the present embodiment, the real patterns formed in the pattern region **200** are line-and-space patterns which are an example of repeating patterns. An example of the pattern region **200** is a memory cell.

(29) First, an image of a first region **201** in the pattern region **200** including patterns formed by the multi-patterning is generated by the scanning electron microscope **50**. After the image of the first region **201** is generated, an image of a second region **202**, which is different from the first region **201**, is generated by the scanning electron microscope **50**. The second region **202** is a region within the pattern region **200**, as well as the first region **201**. Sizes of the first region **201** and the second region **202** are not particularly limited, but are, for example, a size of a field of view (FOV) of the scanning electron microscope **50** or a size of a combination of a plurality of fields of view of the scanning electron microscope **50**.

(30) The first region **201** is a region including a distinctive pattern whose position can be specified in order to ensure that subsequent first matching and second matching can be performed correctly. In the present embodiment, the first region **201** is an edge region including an edge of the pattern region **200**. The second region **202** is located within the pattern region **200** and located more inwardly than the first region **201**.

(31) FIG. **3** is a schematic diagram showing an example of an image **205** of the first region **201**. As shown in FIG. **3**, patterns in the first region **201** include two sets of patterns **210**A and **210**B individually formed by the double patterning. These patterns **210**A and **210**B are arranged alternately. The patterns **210**A and **210**B in the first region **201** are repeating patterns including the edge of the pattern region **200**. Specifically, an outermost pattern **210**A is a distinctive pattern whose position can be specified. Therefore, although these patterns **210**A and **210**B are repeating patterns, accurate pattern matching is ensured. In the following description, the patterns **210**A and **210**B on the image **205** of the first region **201** are referred to as reference patterns. The number of reference patterns **210**A and **210**B included in the first region **201** is not limited to the example shown in FIG. **3**.

(32) As shown in FIG. **3**, the reference patterns **210**A and **210**B do not include a pattern end. Specifically, each of the reference patterns **210**A and **210**B extends across the entire first region **201**. The reason why the reference patterns **210**A and **210**B do not include the pattern end is that pattern shrinkage is likely to occur at the pattern end and a correct pattern width may not be measured. The size and position of the first region **201** can be set manually or automatically. A plurality of first regions **201** may be provided.

(33) Next, as shown in FIG. **4**, the arithmetic system **150** performs the first matching between the plurality of reference patterns **210**A and **210**B on the image **205** of the first region **201** and a plurality of corresponding CAD patterns **215**A and **215**B. The CAD patterns **215**A and **215**B are classified in advance into a first group and a second group according to layer information of these CAD patterns. In the example shown in FIG. **4**, the CAD patterns **215**A are classified into the first group, and the CAD patterns **215**B are classified into the second group. The CAD patterns **215**A and **215**B are created by the arithmetic system **150** based on the design data of the reference patterns **210**A and **210**B.

(34) The first matching is performed as follows. The arithmetic system **150** superimposes one of the SEM image **205** and the CAD patterns **215**A and **215**B on another, and produces gray-level profiles of the SEM image **205** within a range set starting from the edges of the CAD patterns **215**A

and **215**B created from the design data. Then, the arithmetic system **150** determines edges of the reference patterns **210**A and **210**B on the SEM image **205** from the gray-level profiles, and determines a matching position that can minimize bias values between positions of the determined edges and positions of the edges of the corresponding CAD patterns **215**A and **215**B. Each of the bias values is an index value indicating an amount of deviation (or a distance) between an edge determined from each gray-level profile and each edge of the corresponding CAD patterns **215**A and **215**B. The bias values are calculated for all edges in the SEM image **205**.

(35) As shown in FIG. **5**, the arithmetic system **150** classifies the plurality of reference patterns **210**A and **210**B on the SEM image **205** into the first group and the second group according to the layer classification of the CAD patterns **215**A and **215**B. The layer classification of CAD patterns **215**A and **215**B is predetermined based on a pattern-formation order of the multi-patterning (double patterning in this embodiment). In the example shown in FIG. **5**, the arithmetic system **150** classifies the odd-numbered reference patterns **210**A into the first group and the even-numbered reference patterns **210**B into the second group. Depending on the layer classification of the CAD patterns **215**A and **215**B, the arithmetic system **150** may classify the even-numbered reference patterns **210**B into the first group and the odd-numbered reference patterns **210**A into the second group.

(36) Next, the arithmetic system **150** measures widths of the plurality of reference patterns **210**A belonging to the first group and widths of the plurality of reference patterns **210**B belonging to the second group. The measured values of the widths of the reference patterns **210**A and **210**B are CD (Critical Dimension) values of the reference patterns **210**A and **210**B. A process of measuring the widths of the reference patterns **210**A and **210**B is not particularly limited. In one example, the arithmetic system **150** may calculate the sum of an average of the bias values of each reference pattern and a width of the corresponding CAD pattern, and may use the calculated sum as a measured value of the width (i.e., a CD value). In another example, the arithmetic system **150** may measure a distance between peaks of a brightness profile of each reference pattern. In general, a brightness of an edge of a pattern on an image may be higher than those of other parts of the pattern. Therefore, the distance between the peaks of the brightness profile can be used as a measured value of the width of the reference pattern.

(37) The arithmetic system **150** determines a first integrated value by integrating the measured values of the widths of the plurality of reference patterns **210**A belonging to the first group, and determines a second integrated value by integrating the measured values of the widths of the plurality of reference patterns **210**B belonging to the second group.

(38) Further, the arithmetic system **150** compares the first integrated value and the second integrated value, and determines a magnitude relationship between the first integrated value and the second integrated value. Specifically, the arithmetic system **150** determines which of the first integrated value and the second integrated value is larger than the other. The arithmetic system **150** stores the determined magnitude relationship between the first integrated value and the second integrated value in the memory **162**.

(39) The reference patterns **210**A belonging to the first group and the reference patterns **210**B belonging to the second group are real patterns formed by individual steps in the multi-patterning. Usually, in the multi-patterning, widths of real patterns formed by individual steps are slightly different. Therefore, the widths of the reference patterns **210**A belonging to the first group and the widths of the reference patterns **210**B belonging to the second group are also slightly different. The first integrated value calculated for the first group and the second integrated value calculated for the second group reflect the widths of the reference patterns **210**A and **210**B belonging to these two groups.

(40) FIG. **6** is a graph showing a relationship between the number of integrated widths of the reference patterns and the first and second integrated values. As can be seen from FIG. **6**, as the number of integrated widths of the reference patterns **210**A and **210**B increases, the difference

between the first integrated value and the second integrated value increases. The difference between the width of each reference pattern **210**A belonging to the first group and the width of each reference pattern **210**B belonging to the second group is extremely small, but there is a significant difference between the first integrated value and the second integrated value. Therefore, the arithmetic system **150** can determine either the width of the reference pattern **210**A belonging to the first group or the width of the reference pattern **210**B belonging to the second group is larger than the other based on the difference between the first integrated value and the second integrated value. In the example shown in FIG. **6**, the magnitude relationship between the first integrated value and the second integrated value is such that the first integrated value is larger than the second integrated value.

(41) In order to determine the magnitude relationship between the first integrated value and the second integrated value, it is desirable that there is a significant difference between the first integrated value and the second integrated value. From this point of view, the number of integrated reference patterns may be a preset number or more. Alternatively, the arithmetic system **150** may integrate the measured values of the widths of the plurality of reference patterns **210**A belonging to the first group and integrate the measured values of the widths of the plurality of reference patterns **210**B belonging to the second group until an absolute value of the difference between the first integrated value and the second integrated value exceeds a predetermined value. In order to increase the number of reference patterns existing in the first region **201**, the field of view (FOV) may be increased or a plurality of fields of view may be combined.

(42) In general, an outermost pattern in the pattern region **200** may have a width significantly different from other patterns due to the influence of a light proximity effect etc., as shown in FIG. **7**. Therefore, the arithmetic system **150** may calculate the first integrated value and the second integrated value without using at least one reference pattern at the outermost side in the first region **201**. In one embodiment, the arithmetic system **150** may exclude reference pattern(s) having a width exceeding a threshold value from the plurality of reference patterns **210**A and **210**B in the first region **201**, and then calculate the first integrated value and the second integrated value.

(43) The arithmetic system **150** calculates integrated values for patterns in the second region **202** shown in FIG. **2** in the same manner as the reference patterns **210**A and **210**B in the first region **201**. FIG. **8** is a schematic diagram showing an example of an image **225** of the second region **202**. As shown in FIG. **8**, patterns in the second region **202** include two sets of patterns **220**A and **220**B individually formed by the double patterning, as well as the reference patterns **210**A and **210**B in the first region **201**. These patterns **220**A and **220**B are arranged alternately.

(44) The patterns **220**A and **220**B in the second region **202** are repeating patterns that do not include an edge of the pattern region **200** (see FIG. **2**). In one example, the second region **202** is located at the center of the pattern region **200** shown in FIG. **2**. As can be seen from FIG. **8**, the patterns **220**A and **220**B in the second region **202** extend across the entire second region **202** and do not have a distinctive shape that enables a position to be specified. The size and position of the second region **202** can be set manually or automatically. The number of patterns **220**A and **220**B included in the second region **202** is not limited to the example shown in FIG. **8**.

(45) As shown in FIG. **9**, the arithmetic system **150** performs a second matching between the plurality of patterns **220**A and **220**B on the image **225** of the second region **202** and a plurality of corresponding CAD patterns **230**A and **230**B. The CAD patterns **230**A and **230**B are classified in advance into a first group and a second group according to layer information of these CAD patterns. In the example shown in FIG. **9**, the CAD patterns **230**A are classified into the first group, and the CAD patterns **230**B are classified into the second group. Since the second matching is performed in the same manner as the first matching described above, the repetitive descriptions thereof will be omitted.

(46) As shown in FIG. **10**, the arithmetic system **150** classifies the plurality of patterns **220**A and **220**B on the SEM image **225** into the first group and the second group according to the layer

classification of the CAD patterns **230**A and **230**B. In the present embodiment, the arithmetic system **150** classifies the odd-numbered patterns **220**A into the first group and the even-numbered patterns **220**B into the second group. Then, the arithmetic system **150** measures widths of the plurality of patterns **220**A belonging to the first group and widths of the plurality of patterns **220**B belonging to the second group.

(47) The arithmetic system **150** determines a third integrated value by integrating the measured values of the widths of the plurality of patterns **220**A belonging to the first group, and determines a fourth integrated value by integrating the measured values of the widths of the plurality of patterns **220**B belonging to the second group. Furthermore, the arithmetic system **150** compares the third integrated value and the fourth integrated value, and determines a magnitude relationship between the third integrated value and the fourth integrated value. Specifically, the arithmetic system **150** determines which of the third integrated value and the fourth integrated value is larger than the other. The arithmetic system **150** stores the determined magnitude relationship between the third integrated value and the fourth integrated value in the memory **162**.

(48) Similar to the first integrated value and the second integrated value, the number of integrated patterns may be a preset number or more. Alternatively, the arithmetic system **150** may integrate the measured values of the widths of the plurality of patterns **220**A belonging to the first group and integrate the measured values of the widths of the plurality of patterns **220**B belonging to the second group until an absolute value of the difference between the third integrated value and the fourth integrated value exceeds a predetermined value. In order to increase the number of patterns existing in the second region **202**, the field of view (FOV) may be increased or a plurality of fields of view may be combined.

(49) Next, the arithmetic system **150** determines that the second matching is performed correctly when the magnitude relationship between the third integrated value and the fourth integrated value matches the magnitude relationship between the first integrated value and the second integrated value. In the example shown in FIG. **6**, since the first integrated value is larger than the second integrated value, the arithmetic system **150** determines that the magnitude relationship between the third integrated value and the fourth integrated value coincides with the magnitude relationship between the first integrated value and the second integrated value when the third integrated value is larger than the fourth integrated value.

(50) The arithmetic system **150** displays on the display screen **165** the first integrated value, the second integrated value, the third integrated value, the fourth integrated value, the difference between the first integrated value and the second integrated value, the difference between the third integrated value and the fourth integrated value, etc. An operator can visually confirm the results displayed on the display screen **165**.

(51) When the magnitude relationship between the third integrated value and the fourth integrated value does not coincide with the magnitude relationship between the first integrated value and the second integrated value (e.g., when the third integrated value is smaller than the fourth integrated value), it is presumed that the third integrated value is an integrated value of the widths of the plurality of patterns belonging to the second group, and the fourth integrated value is an integrated value of the widths of the plurality of patterns belonging to the first group. Therefore, in this case, the arithmetic system **150** shifts the plurality of patterns **220**A and **220**B on the image **225** of the second region **202** by one pitch relative to the plurality of corresponding CAD patterns **230**A and **230**B, and then performs the second matching again.

(52) According to the present embodiment, accurate pattern matching in the second region **202** can be ensured with reference to the magnitude relationship of the pattern widths in the first region **201**. In particular, according to the present embodiment, the measured values of the pattern widths can be used for optimizing process parameters. In addition, it is possible to monitor pattern widths in a region where a process margin is small.

(53) FIG. **11** is a flowchart illustrating an embodiment of the pattern matching method.

(54) In step 1, the scanning electron microscope **50** generates the image **205** of the first region **201** in the pattern region **200** including the repeating patterns (see FIG. **3**). The image **205** of the first region **201** is sent to the arithmetic system **150**.
(55) In step 2, the arithmetic system **150** performs the first matching between the reference patterns **210**A, **210**B on the image **205** of the first region **201** and the CAD patterns **215**A, **215**B that have been classified in advance into a first group and a second group based on the layer information (see FIG. **4**).
(56) In step 3, the arithmetic system **150** classifies the plurality of reference patterns **210**A and **210**B into the first group and the second group according to the layer classification of the CAD patterns **215**A and **215**B (see FIG. **5**).
(57) In step 4, the arithmetic system **150** measures the widths of the plurality of reference patterns **210**A belonging to the first group and the widths of the plurality of reference patterns **210**B belonging to the second group.
(58) In step 5, the arithmetic system **150** integrates the measured values of the widths of the plurality of reference patterns **210**A belonging to the first group to determine the first integrated value, and integrates the measured values of the widths of the plurality of reference patterns **210**B belonging to the second group to determine the second integrated value.
(59) In step 6, the arithmetic system **150** determines the magnitude relationship between the first integrated value and the second integrated value. Specifically, the arithmetic system **150** determines which of the first integrated value and the second integrated value is larger than the other.
(60) In step 7, the scanning electron microscope **50** generates the image **225** of the second region **202** in the pattern region **200** (see FIG. **8**). The image **225** of the second region **202** is sent to the arithmetic system **150**.
(61) In step 8, the arithmetic system **150** performs the second matching between the patterns **220**A, **220**B on the image **225** of the second region **202** and the CAD patterns **230**A, **230**B that have been classified in advance into a first group and a second group based on the layer information (see FIG. **9**).
(62) In step 9, the arithmetic system **150** classifies the plurality of patterns **220**A and **220**B on the image **225** of the second region **202** into the first group and the second group according to the layer classification of the CAD patterns **230**A and **230**B (see FIG. **10**).
(63) In step 10, the arithmetic system **150** measures the widths of the plurality of patterns **220**A belonging to the first group and the widths of the plurality of patterns **220**B belonging to the second group.
(64) In step 11, the arithmetic system **150** integrates the measured values of the widths of the plurality of patterns **220**A belonging to the first group to determine the third integrated value, and integrates the measured values of the widths of the plurality of patterns **220**B belonging to the second group to determine the fourth integrated value.
(65) In step 12, the arithmetic system **150** determines the magnitude relationship between the third integrated value and the fourth integrated value.
(66) In step 13, the arithmetic system **150** compares the magnitude relationship between the third integrated value and the fourth integrated value with the magnitude relationship between the first integrated value and the second integrated value, and determines whether the second matching has been correctly performed based on the comparison result. Specifically, the arithmetic system **150** determines that the second matching has been performed correctly when the magnitude relationship between the third integrated value and the fourth integrated value coincides with the magnitude relationship between the first integrated value and the second integrated value. When the magnitude relationship between the third integrated value and the fourth integrated value does not coincide with the magnitude relationship between the first integrated value and the second integrated value, the arithmetic system **150** shifts the plurality of patterns **220**A and **220**B on the image **225** of the second region **202** by one pitch relative to the plurality of corresponding CAD

patterns **230**A and **230**B, and then performs the second matching again.

(67) In the above-described embodiments, the repeating patterns are formed by the double patterning, but the present invention is not limited to the above-described embodiments. The present invention is also applicable to repeating patterns formed by other multi-patterning, such as quadruple patterning. For example, patterns formed by the quadruple patterning are classified into four groups according to layer classification of corresponding CAD patterns, four integrated values are calculated for these four groups, and a magnitude relation of these four integrated values is determined.

(68) Furthermore, the present invention can be applied to repeating patterns other than the line and space patterns. FIG. **12** is a schematic diagram showing hole patterns on an image **305** of a first region **201**. The hole patterns are an example of repeating patterns formed by the double patterning. In the example shown in FIG. **12**, the hole patterns on the image **305** includes two sets of hole patterns **310**A, **310**B individually formed by the double patterning. These patterns **310**A and **310**B are arranged alternately in the X direction and the Y direction. In the following descriptions, these hole patterns **310**A and **310**B will be referred to as reference patterns. In order to specify the positions of the reference patterns **310**A and **310**B in the X direction and the Y direction, the first region **201** is an edge region and a corner region of the pattern region **200** shown in FIG. **2**.

(69) The arithmetic system **150** classifies the reference patterns **310**A and **310**B into a first group and a second group according to layer classification of corresponding CAD patterns. In one embodiment, the arithmetic system **150** classifies the first reference pattern **310**A into the first group and the second reference pattern **310**B into the second group.

(70) FIG. **13** is a schematic diagram showing hole patterns on an image **325** of a second region **205**. As shown in FIG. **13**, the patterns constituted of the hole patterns in the second region **202** include two sets of patterns **320**A and **320**B individually formed by the double patterning. These patterns **320**A and **320**B are arranged alternately in the X direction and the Y direction. The arithmetic system **150** classifies the hall patterns **320**A and **320**B into a first group and a second group according to layer classification of corresponding CAD patterns. In one embodiment, the arithmetic system **150** classifies the hall pattern **320**A into the first group and the ball pattern **320**B into the second group.

(71) The pattern matching of this embodiment is performed in the same manner as the embodiment described with reference to FIG. **11**.

(72) In step 1, the scanning electron microscope **50** generates the image **305** of the first region **201** in the pattern region **200** including the repeating patterns (see FIG. **12**). The image **305** of the first region **201** is sent to the arithmetic system **150**.

(73) In step 2, the arithmetic system **150** performs the first matching between the reference patterns **310**A, **310**B on the image **305** of the first region **201** and the CAD patterns that have been classified in advance into a first group and a second group based on the layer information.

(74) In step 3, the arithmetic system **150** classifies the plurality of reference patterns **310**A and **310**B into the first group and the second group according to the layer classification of the corresponding CAD patterns.

(75) In step 4, the arithmetic system **150** measures widths of the plurality of reference patterns **310**A belonging to the first group and widths of the plurality of reference patterns **310**B belonging to the second group.

(76) In step 5, the arithmetic system **150** integrates the measured values of the widths of the plurality of reference patterns **310**A belonging to the first group to determine a first integrated value, and integrates the measured values of the widths of the plurality of reference patterns **310**B belonging to the second group to determine a second integrated value.

(77) In step 6, the arithmetic system **150** determines a magnitude relationship between the first integrated value and the second integrated value. Specifically, the arithmetic system **150** determines which of the first integrated value and the second integrated value is larger than the other.

(78) In step 7, the scanning electron microscope **50** generates the image **325** of the second region **202** in the pattern region **200** (see FIG. **13**). The image **325** of the second region **202** is sent to the arithmetic system **150**.
(79) In step 8, the arithmetic system **150** performs the second matching between the patterns **320**A, **320**B on the image **325** of the second region **202** and the CAD patterns that have been classified in advance into a first group and a second group based on the layer information.
(80) In step 9, the arithmetic system **150** classifies the plurality of patterns **320**A and **320**B on the image **325** of the second region **202** into the first group and the second group according to the layer classification of the corresponding CAD patterns.
(81) In step 10, the arithmetic system **150** measures widths of the plurality of patterns **320**A belonging to the first group and widths of the plurality of patterns **320**B belonging to the second group.
(82) In step 11, the arithmetic system **150** integrates the measured values of the widths of the plurality of patterns **320**A belonging to the first group to determine a third integrated value, and integrates the measured values of the widths of the plurality of patterns **320**B belonging to the second group to determine a fourth integrated value.
(83) In step 12, the arithmetic system **150** determines a magnitude relationship between the third integrated value and the fourth integrated value.
(84) In step 13, the arithmetic system **150** compares the magnitude relationship between the third integrated value and the fourth integrated value with the magnitude relationship between the first integrated value and the second integrated value, and determines whether the second matching has been correctly performed based on the comparison result. Specifically, the arithmetic system **150** determines that the second matching has been performed correctly when the magnitude relationship between the third integrated value and the fourth integrated value coincides with the magnitude relationship between the first integrated value and the second integrated value. When the magnitude relationship between the third integrated value and the fourth integrated value does not coincide with the magnitude relationship between the first integrated value and the second integrated value, the arithmetic system **150** shifts the plurality of patterns on the image **325** of the second region **202** by one pitch relative to the plurality of corresponding CAD patterns, and then performs the second matching again.
(85) In the embodiments described previously, the integrated value of the widths of the patterns belonging to each group is calculated. Instead of the widths of the patterns, an integrated value of slopes of brightness profiles of edges of patterns on an image may be calculated. Usually, in the multi-patterning, slopes of edges of real patterns formed by individual steps may be slightly different due to an etching process. Therefore, the slopes of the brightness profiles of the edges of the patterns appearing on the image are also slightly different between groups.
(86) FIG. **14** is a diagram showing an example of a brightness profile of a pattern on an image. The brightness profile of the pattern is a distribution of brightness along a direction across the pattern. The brightness is represented by, for example, a numerical value ranging from 0 to 255 according to a gray scale. As can be seen from FIG. **14**, a slope S of the brightness profile of the edge of the pattern can be calculated from brightness values and a distance (number of pixels).
(87) Hereinafter, in the examples shown in FIGS. **2** to **5**, an embodiment in which the slope of the brightness profile of the edge of the pattern is used instead of the width of the pattern will be described. Since the details of the following embodiments, which will not be particularly described, are the same as the details of the embodiments described with reference to FIGS. **2** to **13**, the duplicated descriptions thereof will be omitted.
(88) FIG. **15** is a flowchart illustrating an embodiment of a pattern matching method using the slope of the brightness profile of the pattern edge.
(89) In step 2-1, the scanning electron microscope **50** generates the image **205** of the first region **201** in the pattern region **200** including the repeating patterns (see FIG. **3**). The image **205** of the

first region **201** is sent to the arithmetic system **150**.

(90) In step 2-2, the arithmetic system **150** performs the first matching between the reference patterns **210**A, **210**B on the image **205** of the first region **201** and the CAD patterns **215**A, **215**B that have been classified in advance into a first group and a second group based on the layer information (see FIG. **4**).

(91) In step 2-3, the arithmetic system **150** classifies the plurality of reference patterns **210**A and **210**B into the first group and the second group according to the layer classification of the CAD patterns **215**A and **215**B (see FIG. **5**).

(92) In step 2-4, the arithmetic system **150** calculates slopes of brightness profiles of the edges of the plurality of reference patterns **210**A belonging to the first group and slopes of brightness profiles of the edges of the plurality of reference patterns **210**B belonging to the second group.

(93) In step 2-5, the arithmetic system **150** integrates the calculated values of the slopes of the brightness profiles of the edges of the plurality of reference patterns **210**A belonging to the first group to determine a first integrated value, and integrates the calculated values of the slopes of the brightness profiles of the edges of the plurality of reference patterns **210**B belong to the second group to determine a second integrated value.

(94) The difference between the slope of the brightness profile of the edge of each reference pattern **210**A belonging to the first group and the slope of the brightness profile of the edge of each reference pattern **210**B belonging to the second group is extremely small, but there is a significant difference between the first integrated value and the second integrated value. In step 2-6, the arithmetic system **150** determines a magnitude relationship between the first integrated value and the second integrated value. Specifically, the arithmetic system **150** determines which of the first integrated value and the second integrated value is larger than the other.

(95) In step 2-7, the scanning electron microscope **50** generates the image **225** of the second region **202** in the pattern region **200** (see FIG. **8**). The image **225** of the second region **202** is sent to the arithmetic system **150**.

(96) In step 2-8, the arithmetic system **150** performs the second matching between the patterns **220**A, **220**B on the image **225** of the second region **202** and the CAD patterns **230**A, **230**B that have been classified in advance into a first group and a second group based on the layer information (see FIG. **9**).

(97) In step 2-9, the arithmetic system **150** classifies the plurality of patterns **220**A and **220**B on the image **225** of the second region **202** into the first group and the second group according to the layer classification of the CAD patterns **230**A and **230**B (see FIG. **10**).

(98) In step 2-10, the arithmetic system **150** calculates slopes of brightness profiles of edges of the plurality of patterns **220**A belonging to the first group and slopes of brightness profiles of edges of the plurality of patterns **220**B belonging to the second group.

(99) In step 2-11, the arithmetic system **150** integrates the calculated values of the slopes of the brightness profiles of the edges of the plurality of patterns **220**A belonging to the first group to determine a third integrated value, and integrates the calculated values of the slopes of the brightness profiles of the edges of the plurality of patterns **220**B belonging to the second group to determine a fourth integrated value.

(100) In step 2-12, the arithmetic system **150** determines a magnitude relationship between the third integrated value and the fourth integrated value.

(101) In step 2-13, the arithmetic system **150** compares the magnitude relationship between the third integrated value and the fourth integrated value with the magnitude relationship between the first integrated value and the second integrated value, and determines whether the second matching has been correctly performed based on the comparison result. Specifically, the arithmetic system **150** determines that the second matching has been performed correctly when the magnitude relationship between the third integrated value and the fourth integrated value coincides with the magnitude relationship between the first integrated value and the second integrated value. When

the magnitude relationship between the third integrated value and the fourth integrated value does not coincide with the magnitude relationship between the first integrated value and the second integrated value, the arithmetic system **150** shifts the plurality of patterns **220**A and **220**B on the image **225** of the second region **202** by one pitch relative to the plurality of corresponding CAD patterns **230**A and **230**B, and then performs the second matching again.
(102) Although detailed descriptions are omitted, the embodiment described with reference to FIG. **15** can be applied to the hole patterns shown in FIGS. **12** and **13** as well.
(103) The previous description of embodiments is provided to enable a person skilled in the art to make and use the present invention. Moreover, various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles and specific examples defined herein may be applied to other embodiments. Therefore, the present invention is not intended to be limited to the embodiments described herein but is to be accorded the widest scope as defined by limitation of the claims.
INDUSTRIAL APPLICABILITY
(104) The present invention is applicable to a method of performing matching between a pattern formed on a surface of a workpiece, such as a wafer or a glass substrate, used for manufacturing semiconductor devices and a CAD pattern created from pattern design data.
REFERENCE SIGNS LIST
(105) **50** scanning electron microscope **111** electron gun **112** converging lens **113** X deflector **114** Y deflector **115** objective lens **116** lens controller **117** deflection controller **118** image acquisition device **120** specimen chamber **121** specimen stage **122** stage controller **124** wafer **130** secondary electron detector **131** backscattered electron detector **140** transfer device **150** arithmetic system **161** database **162** memory **163** processor **165** display screen **200** pattern region **201** first region **202** second region **205** image **210**A, **210**B reference pattern **220**A, **220**B pattern **225** image

Claims

1. A pattern matching method comprising: generating an image of a first region in a pattern region containing patterns formed by multi-patterning; performing first matching between a plurality of reference patterns on the image of the first region and a plurality of first CAD patterns that have been classified in advance into a first group and a second group according to layer information; classifying the plurality of reference patterns into the first group and the second group according to the layer classification of the first CAD patterns; measuring widths of the plurality of reference patterns belonging to the first group and widths of the plurality of reference patterns belonging to the second group; determining a first integrated value by integrating measured values of the widths of the plurality of reference patterns belonging to the first group; determining a second integrated value by integrating measured values of the widths of the plurality of reference patterns belonging to the second group; determining a magnitude relationship between the first integrated value and the second integrated value; generating an image of a second region in the pattern region; performing second matching between a plurality of patterns on the image of the second region and a plurality of second CAD patterns that have been classified in advance into a first group and a second group according to layer information; classifying the plurality of patterns on the image of the second region into the first group and the second group according to the layer classification of the second CAD patterns; measuring widths of the plurality of patterns belonging to the first group and widths of the plurality of patterns belonging to the second group; determining a third integrated value by integrating measured values of the widths of the plurality of patterns belonging to the first group; determining a fourth integrated value by integrating measured values of the widths of the plurality of patterns belonging to the second group; determining a magnitude relationship between the third integrated value and the fourth integrated value; determining that the second matching has been performed correctly when the magnitude relationship between the third integrated value and

the fourth integrated value coincides with the magnitude relationship between the first integrated value and the second integrated value.

2. The pattern matching method according to claim 1, wherein the first region is an edge region including an edge of the pattern region, and the second region is within the pattern region and is located more inwardly than the first region.

3. The pattern matching method according to claim 1, wherein the patterns formed by the multi-patterning are repeating patterns.

4. The pattern matching method according to claim 1, further comprising: shifting the plurality of patterns on the image of the second region relative to the plurality of second CAD patterns by one pitch when the magnitude relationship between the third integrated value and the fourth integrated value does not coincide with the magnitude relationship between the first integrated value and the second integrated value; and then performing the second matching again.

5. The pattern matching method according to claim 1, wherein an absolute value of a difference between the first integrated value and the second integrated value and an absolute value of a difference between the third integrated value and the fourth integrated value are larger than a predetermined value.

6. A pattern matching method comprising: generating an image of a first region in a pattern region containing patterns formed by multi-patterning; performing first matching between a plurality of reference patterns on the image of the first region and a plurality of first CAD patterns that have been classified in advance into a first group and a second group according to layer information; classifying the plurality of reference patterns into the first group and the second group according to the layer classification of the first CAD patterns; calculating slopes of brightness profiles of edges of the plurality of reference patterns belonging to the first group and slopes of brightness profiles of edges of the plurality of reference patterns belonging to the second group; determining a first integrated value by integrating calculated values of the slopes of the brightness profiles of the edges of the plurality of reference patterns belonging to the first group; determining a second integrated value by integrating calculated values of the slopes of the brightness profiles of the edges of the plurality of reference patterns belonging to the second group; determining a magnitude relationship between the first integrated value and the second integrated value; generating an image of a second region in the pattern region; performing second matching between a plurality of patterns on the image of the second region and a plurality of second CAD patterns that have been classified in advance into a first group and a second group according to layer information; classifying the plurality of patterns on the image of the second region into the first group and the second group according to the layer classification of the second CAD patterns; calculating slopes of brightness profiles of edges of the plurality of patterns belonging to the first group and slopes of brightness profiles of edges of the plurality of patterns belonging to the second group; determining a third integrated value by integrating calculated values of the slopes of the brightness profiles of the edges of the plurality of patterns belonging to the first group; determining a fourth integrated value by integrating calculated values of the slopes of the brightness profiles of the edges of the plurality of patterns belonging to the second group; determining a magnitude relationship between the third integrated value and the fourth integrated value; determining that the second matching has been performed correctly when the magnitude relationship between the third integrated value and the fourth integrated value coincides with the magnitude relationship between the first integrated value and the second integrated value.

7. The pattern matching method according to claim 6, wherein the first region is an edge region including an edge of the pattern region, and the second region is within the pattern region and is located more inwardly than the first region.

8. The pattern matching method according to claim 6, wherein the patterns formed by the multi-patterning are repeating patterns.

9. The pattern matching method according to claim 6, further comprising: shifting the plurality of

patterns on the image of the second region relative to the plurality of second CAD patterns by one pitch when the magnitude relationship between the third integrated value and the fourth integrated value does not coincide with the magnitude relationship between the first integrated value and the second integrated value; and then performing the second matching again.

10. The pattern matching method according to claim 6, wherein an absolute value of a difference between the first integrated value and the second integrated value and an absolute value of a difference between the third integrated value and the fourth integrated value are larger than a predetermined value.